What's New in AI: A 2026 Recap

D-Lab AI Pulse Workshop

Materials: https://github.com/dlab-berkeley/AI_Pulse

Feedback form: https://forms.gle/x8KfejgCViE7t6iQA

Today's Roadmap

1. What happened in 2026
   - Model Context Protocol and customizable AI
   - Agents that act on their own
   - AI starts doing research
   - Big tech eats every industry
2. What's happening right now
   - Models too powerful to release
   - AI is proving math
   - How is AI affecting society?
3. What does the future look like
   - The centralization question
   - Beyond LLMs

MCP: Giving Tools to Your AI

- Initially, AI was stuck in the chat window
- You had to continuously copy-paste, update documents, and share information
- The solution: Model Context Protocol, now the industry standard
- "USB-C for AI": any LLM, any tool; reads files, runs queries, takes actions
- Setup: thousands of ready-made MCP servers; custom ones built in a few hundred lines with the SDK
- Introduced by Anthropic (late 2024); now broadly supported across Claude, Cursor, VS Code, and others

What You Can Do With MCP

Power your AI (read)

- Zotero (citations, PDFs)
- Google Drive / OneDrive / Box
- Notion / Obsidian
- GitHub
- PostgreSQL / Snowflake
- Local files and PDFs

Let your AI act (write)

- Slack: send and search messages
- Asana / Linear / Jira: file tickets
- Gmail / Calendar: draft, schedule
- Figma: produce design drafts
- Hex and spreadsheets: analyze
- Browser (Playwright): navigate

Customize and Share Your AI

- Three platforms, one idea: customize the AI, share with a link
- Behavior packaging: Skills (Claude), Gems (Gemini), Custom GPTs (ChatGPT)
- Interactive apps: Artifacts (Claude), Canvas (ChatGPT and Gemini)
- With Skills: format a manuscript, build slides from a dataset, run code reviews
- With Artifacts: build a live dashboard, simulate a physics setup, visualize math, prototype a teaching tool

From Chats to Tasks

- Chats: you ask, it answers
- Tasks: you describe an outcome; the AI plans, acts, iterates
- Reads your files, browses the web, runs code, edits documents
- Works for minutes to hours; step away, come back to results
- Amplifies effectiveness, but also amplifies risks

Claude Code and Gemini CLI

- AI agents that live in your terminal
- Read and edit your codebase directly
- Run commands, scripts, and tests
- Know your file structure and Git history
- Spawn subagents to work in parallel
- Claude Code: steerable from your phone (Remote Control)

Cowork

- AI that runs on your desktop
- Reads and creates files (PDFs, Word, Excel, slides)
- Organizes folders, builds tables, drafts documents
- Connects to apps via MCP (Slack, Notion, Drive)
- Computer Use: controls any app on your screen (macOS research preview)
- Take a task, walk away, come back to results
- Schedule recurring jobs (daily arXiv, weekly reports)

Claude in Chrome (and Project Mariner, ChatGPT Agent)

- AI that drives your browser
- Sees pages, clicks, fills forms, navigates tabs
- Library portals, journal sites, IRB forms, grant submissions
- Faster than full screen control for web-only tasks
- Equivalents: Project Mariner (Google) and ChatGPT Agent (OpenAI)

OpenClaw

- Open-source agent in your messaging apps (Signal, Telegram, WhatsApp, Discord)
- 100+ skills; model-agnostic; runs locally on your machine
- Common uses: morning briefings, email triage, coding from your phone
- Power workflows: parallel research, smart home via chat, self-healing servers
- Viral arc: 250K stars in 60 days; creator joined OpenAI Feb 2026
- Caveat: Cisco found 12% of community skills compromised
- Highlights growing security risks in open agent ecosystems

Automated Research: Agents Doing Science

- Agents can read your files and act on your tools
- The next question: can they do science?
- Research is a loop: hypothesize, experiment, analyze, write, review
- In 2026, every step has a working AI tool
- Important caveat: AI-generated papers and experiments still require rigorous human verification
- Typical cost: tens to hundreds of dollars per full research run

Karpathy's AutoResearch

- Andrej Karpathy, released March 2026
- ~630 lines of Python; tens of thousands of GitHub stars in days
- AI agent runs a small ML training setup
- Modifies code, trains 5 minutes, scores result, keeps or discards, repeats
- 700 experiments in 2 days; 20 optimizations discovered
- Karpathy's term: "the self-improvement loopy era"

`github.com/karpathy/autoresearch`

Sakana AI Scientist

- End-to-end agent: idea, code, experiments, paper, peer review
- First AI-generated paper to pass double-blind peer review at the ICLR 2025 ICBINB workshop; later withdrawn as part of the experiment
- Paper: "Compositional Regularization: Unexpected Obstacles in Enhancing Neural Network Generalization"
- Sakana team's human-authored Nature paper (March 2026) describes the system and documents a clear scaling law: better base models produce dramatically better AI-generated papers

`github.com/SakanaAI/AI-Scientist-v2` `sakana.ai/ai-scientist-first-publication`

HKU AI-Researcher

- NeurIPS 2025 Spotlight; University of Hong Kong
- Reads relevant literature
- Generates novel research ideas
- Codes and runs experiments
- Analyzes results and writes the manuscript
- Live demo at novix.science

`github.com/HKUDS/AI-Researcher`

refine.ink

- AI as an obsessive peer reviewer
- Founded 2025 by Benjamin Golub (Northwestern) and Yann Calvo Lopez (CNRS)
- Reads finished drafts; flags errors, inconsistencies, logical flaws
- 30-minute turnaround using hundreds of parallel LLM calls
- Used at Harvard, Stanford, Yale, MIT, Caltech
- Closed source; first review free

How Anthropic Uses Its Own Tools

> *"I'm Boris and I created Claude Code. Lots of people have asked how I use Claude Code..."*
>
> *"I run 5 Claudes in parallel in my terminal. I number my tabs 1-5, and use system notifications to know when a Claude needs input."*
>
> *"I also run 5-10 Claudes on claude.ai/code, in parallel with my local Claudes. I also start a few sessions from my phone every morning and throughout the day."*

Boris Cherny (creator of Claude Code), January 2, 2026

Anthropic Mythos

- Anthropic's newest frontier model: a step up in general capability
- Particularly strong at finding software flaws (downstream of better coding and reasoning)
- So powerful Anthropic delayed public release over dual-use concerns
- Project Glasswing: access for security companies first, so defenders can harden systems
- Found vulnerabilities in FFmpeg, Firefox, and FreeBSD

The Supply Chain Twist

- Mar 3, 2026: DoD designates Anthropic a supply-chain risk; Pentagon blocked from Mythos
- The NSA is using Mythos anyway
- Apr 16: White House moves to give other US agencies access (Bloomberg)
- Banks pursuing access: Goldman, Citi, BofA, Morgan Stanley
- China's 360 Digital Security has a parallel tool (∼1,000 unknown bugs found)
- Google M-Trends 2026: mean time to exploit has gone negative

AI Can Now Reason

- The big unlock of 2026: AI is doing real mathematical reasoning
- Not just generating plausible-looking proofs
- Actually closing open problems
- Math is the cleanest test: a proof either works or it doesn't
- Speed of discoveries is accelerating

From One Problem to One a Day

- Oct 2025: GPT-5 helps Sawhney + Sellke rediscover forgotten 2003 solution to Erdős #339 via superior literature retrieval
- Jan 2026: First fully autonomous Lean proof (Aristotle agent, #728)
- Apr 9: OpenAI internal model closes 5 problems in a single batch
- Apr 24, 2026: Liam Price solves Erdős #1196 in a single prompt (raw output required significant expert refinement)
- Apr 25, 2026: Additional progress on Erdős #1138 using GPT-5.5

AI as Math Collaborator

- AI is now part of math research at every level: experts and amateurs alike
- Mehtaab Sawhney (Columbia) and Mark Sellke (Harvard): used GPT-5 to surface a forgotten 2003 result and crack Erdős #339
- Liam Price (23-year-old amateur): cracked Erdős #1196 in a single ChatGPT prompt
- Pattern: AI proposes ideas and partial proofs; humans formalize and verify
- Tao: “The job description is changing”

https://chatgpt.com/share/69dd1c83-b164-8385-bf2e-8533e9baba9c

How Is AI Affecting Society?

- Beyond the labs, AI is reshaping society in real time
- The loudest debate: jobs and the labor market
- Quieter but big: defense entanglement, community pushback, energy demand

Dario Amodei's Warning

- Anthropic CEO; Axios interview, May 2025
- Up to 50% of entry-level white-collar jobs could disappear
- US unemployment to 10-20% in 1-5 years
- Sectors hit hardest: tech, finance, law, consulting
- "Cancer is cured, the economy grows at 10% a year, the budget is balanced, and 20% of people don't have jobs."
- Continued to warn strongly throughout 2026, including at Davos; software engineers replaced in 6-12 months

Companies Already Cutting

- Salesforce (Benioff, Jan 2025): "maybe we aren't going to hire anybody this year"; cited 30%+ engineering productivity
- Klarna (May 2025): reversed its AI-only customer-service push after quality dropped
- Wall Street Q1 2026: 6 big banks cut 5,000 net (15,000 gross) while booking $47B profit
- Big Tech Q1 2026: Meta + Microsoft ~20,000 combined; Amazon ~30,000 since Oct 2025
- AI cited in 25% of March 2026 layoff announcements (Challenger, Gray & Christmas)

The Dissenters

- Yann LeCun (April 2026): "Dario is wrong. He knows absolutely nothing about the effects of technological revolutions on the labor market."
- LinkedIn (April 15): hiring down 20% since 2022, but no measurable AI displacement signal in support, admin, or marketing
- Andrew Ng (Davos 2026): "AI can only do 30-40% of the work now and for the foreseeable future"
- Daron Acemoglu (MIT, Nobel 2024): only ~20% of tasks AI-exposed; modest productivity gains
- Sundar Pichai (Cloud Next 2026): augmentation-positive; "every employee can become a builder"

Beyond Jobs: AI's Broader Footprint

- Anthropic Institute (March 2026) houses the Societal Impacts team; OpenAI's parallel is its Global Affairs and Policy team
- Pentagon-Anthropic row: DoD designated Anthropic a supply-chain risk (March 2026); federal judge blocked the designation
- Palantir defense: Maven targeting in Iran; ImmigrationOS for ICE deportations; $300M USDA food-supply deal (April 22)
- Communities pushing back: Maine's legislature passed a statewide data center moratorium (later vetoed by governor)
- Utah: Kevin O'Leary's 9 GW Wonder Valley campus approved, projected to consume more than twice the state's current electricity

Sam Altman: Centralization Could Lead to Ruin

- Feb 19, 2026 (India AI Impact Summit): "Centralisation of this technology in one company or country could lead to ruin"
- Called for an IAEA-style global AI regulator
- Warned superintelligence could arrive "within a couple of years"
- "The Gentle Singularity" (June 2025): "Make superintelligence cheap, widely available, and not too concentrated"
- The tension: OpenAI is simultaneously building "a gigawatt of new AI infrastructure every week" (BlackRock Summit, March 2026)

Yann LeCun: LLMs Are a Dead End

- "LLMs are an off-ramp on the road to human-level AI"
- Nov 2025: left Meta after 12 years (FAIR's founding director)
- March 2026: founded AMI Labs (Advanced Machine Intelligence)
- Raised $1.03B at $3.5B pre-money
- His architecture: JEPA, predicting representations rather than tokens
- He's not alone

Fei-Fei Li and World Models

- Founded World Labs in 2024 with Justin Johnson, Christoph Lassner, Ben Mildenhall
- Thesis: "spatial intelligence"; AI must understand 3D space, physics, and persistent geometry
- Marble (launched Nov 2025): generates persistent, editable 3D environments from text, images, video
- Feb 2026: $1B from AMD, Autodesk, NVIDIA, Fidelity; in talks at $5B valuation
- The bet: language alone isn't enough; physical understanding is the unlock

Ineffable Intelligence

- Founded by David Silver (DeepMind, architect of AlphaGo and AlphaZero)
- April 27, 2026 (yesterday): $1.1B seed at $5.1B valuation
- Sequoia, NVIDIA, UK government; largest European seed ever
- Thesis: a “superlearner” that learns from environmental interaction with no pre-training on human data
- Foundation paper: Silver and Richard Sutton (2024 ACM Turing Award co-recipient), “Welcome to the Era of Experience”

What They Promise

- Predict how the world changes after an action (physical reasoning)
- Robotics and embodied control (Gemini Robotics, NVIDIA Cosmos)
- Persistent 3D worlds (Marble, DeepMind's Genie 3 at 24 fps)
- Discovery beyond human data (AlphaProof's IMO silver via self-play)
- Three Turing-adjacent figures betting against pure LLM scaling